Python Automata Library (PAL) Cheat Sheet

import PythonAutomataLibrary as pal · Safe mode is default. Call **pal.FastMode**() before constructing any PAL object. Put substantive model kernels under **@pal.njit(cache=True)**.

Factories and utilities

FastMode() — Select PAL fast mode before...
Seed(seed) — Seed PAL's shared random-number...
Random() — Draw a uniform random value from...
RandInt(max) — Draw an integer from PAL's random...
AwaitWindows(timeout=30) — Wait for PAL windows that are still...
ColorScale(colors, value) — Map a scalar value to an RGB color...
NewIList() — Create an empty PAL integer list...
MooreHood(dim, excludeCenter=False) — Build relative integer offsets for a...
VonNeumannHood(dim, excludeCenter=False) — Build relative integer offsets for a...
CircleHood(dim, rad, excludeCenter=False) — Build relative integer offsets...
NewMultinomial(other=None) — Create a PAL multinomial/binomial...
NewGrid(dimensions, dtype) — Create a typed lattice grid with the...
NewAgentGrid(dimensions, numAgentProps=0, isStackable=False) — Create a grid for individual agents;...
NewPopGrid(dimensions, capacity=None) — Create a lattice of integer...
NewPDEgrid(dimensions) — Create a lattice-backed scalar field...

Multinomial

Binomial(n: int, p: float) -> int — Draw a binomial sample with n trials...
Setup(n: int) — Configure the sampler for n trials...
Sample(p: float) -> int — Draw a binomial sample using the...

IList

__len__() -> int — Return the number of integers...
__getitem__(index: int) -> int — Return the integer stored at the...
Append(i: int) — Append integer i to the end of the list.
Clear() — Remove all values from the list.
Random() -> int — Return a randomly selected value...
Shuffle() — Shuffle the list in place using...
All() -> np.ndarray — Return a detached NumPy array...
Iter() -> np.ndarray — Iterate over the current list values...

AgentGrid

__len__() -> int — Return the number of living agents...
xDim() -> int — Number of lattice sites along x.
yDim() -> int — Number of lattice sites along y;...
zDim() -> int — Number of lattice sites along z;...
nDims() -> int — Number of spatial dimensions.
wrapX() -> bool — Whether x uses periodic boundary...
wrapY() -> bool — Whether y uses periodic boundary...
wrapZ() -> bool — Whether z uses periodic boundary...
Box(x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None, z2: int=None) — Iterate lattice coordinates in the...
Hood(hood, x: int, y: int=None, z: int=None, *, unroll: bool=False) — Apply relative neighborhood offsets...
GetPop() -> int — Return the number of living agents...
Alive(agent: int) -> bool — Return whether agent currently...
NewAgentSQ(x: int, y: int=-1, z: int=-1) -> int — Create an agent at the center of a...
NewAgent(x: float, y: float=-1.0, z: float=-1.0) -> int — Create an agent at the supplied...
Dispose(agent: int) — Remove a living agent from the grid;...
I(agent: int) -> int — Return the linear lattice-site index...
XSQ(agent: int) -> int — Return the agent's integer x site...
YSQ(agent: int) -> int — Return the agent's integer y site...
ZSQ(agent: int) -> int — Return the agent's integer z site...
X(agent: int) -> float — Return the agent's continuous x...
Y(agent: int) -> float — Return the agent's continuous y...
Z(agent: int) -> float — Return the agent's continuous z...
__getitem__(key) -> float — Read a supported per-agent property...
__setitem__(key, value: float) — Set a supported per-agent property...
ToI(x: int, y: int=-1, z: int=-1) -> int — Convert lattice coordinates to the...
ItoX(i: int) -> int — Convert a linear site index to its x...
ItoY(i: int) -> int — Convert a linear site index to its y...

AgentGrid

ItoZ(i: int) -> int — Convert a linear site index to its z...
InWrapSQX(value: int) -> int — Wrap an integer x site coordinate...
InWrapSQY(value: int) -> int — Wrap an integer y site coordinate...
InWrapSQZ(value: int) -> int — Wrap an integer z site coordinate...
InWrapX(value: float) -> float — Wrap a continuous x coordinate when...
InWrapY(value: float) -> float — Wrap a continuous y coordinate when...
InWrapZ(value: float) -> float — Wrap a continuous z coordinate when...
DispWrapX(x1: float, x2: float) -> float — Return the shortest signed x...
DispWrapY(y1: float, y2: float) -> float — Return the shortest signed y...
DispWrapZ(z1: float, z2: float) -> float — Return the shortest signed z...
counts() -> object — Read-only occupancy counts for...
LastAgent(x: int, y: int=-1, z: int=-1) -> int — Return the last agent handle at the...
MoveSQ(agent: int, x: int, y: int=-1, z: int=-1) — Move an agent to a lattice site...
Move(agent: int, x: float, y: float=-1.0, z: float=-1.0) — Move an agent to continuous...
AgentsAt(x: int, y: int=-1, z: int=-1) — Return the agent handle or handles...
AgentsInRadius(rad: float, x: float, y: float=-1.0, z: float=-1.0, *, exclude: int=None) — Return agents within radius rad of a...
All(shuffle: bool=False) -> np.ndarray — Return a snapshot of living agent...

Grid

__len__() -> int — Return the number of lattice sites.
__getitem__(key) — Read the value or state selected by...
__setitem__(key, value) — Assign the value or state selected...
xDim() -> int — Number of lattice sites along x.
yDim() -> int — Number of lattice sites along y;...
zDim() -> int — Number of lattice sites along z;...
nDims() -> int — Number of spatial dimensions.
wrapX() -> bool — Whether x uses periodic boundary...
wrapY() -> bool — Whether y uses periodic boundary...
wrapZ() -> bool — Whether z uses periodic boundary...
Box(x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None, z2: int=None) — Iterate lattice coordinates in the...
Hood(hood, x: int, y: int=None, z: int=None, *, unroll: bool=False) — Apply relative neighborhood offsets...
ToI(x: int, y: int=-1, z: int=-1) -> int — Convert lattice coordinates to a...
ItoX(i: int) -> int — Convert a linear site index to its x...
ItoY(i: int) -> int — Convert a linear site index to its y...
ItoZ(i: int) -> int — Convert a linear site index to its z...
InWrapX(value: int) -> int — Wrap an x coordinate when x is...
InWrapY(value: int) -> int — Wrap a y coordinate when y is...
InWrapZ(value: int) -> int — Wrap a z coordinate when z is...

PopGrid

__len__() -> int — Return the number of lattice sites.
xDim() -> int — Number of lattice sites along x.
yDim() -> int — Number of lattice sites along y;...
zDim() -> int — Number of lattice sites along z;...
nDims() -> int — Number of spatial dimensions.
wrapX() -> bool — Whether x uses periodic boundary...
wrapY() -> bool — Whether y uses periodic boundary...
wrapZ() -> bool — Whether z uses periodic boundary...
Box(x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None, z2: int=None) — Iterate lattice coordinates in the...
Hood(hood, x: int, y: int=None, z: int=None, *, unroll: bool=False) — Apply relative neighborhood offsets...
ToI(x: int, y: int=-1, z: int=-1) -> int — Convert lattice coordinates to a...
ItoX(i: int) -> int — Convert a linear site index to its x...
ItoY(i: int) -> int — Convert a linear site index to its y...
ItoZ(i: int) -> int — Convert a linear site index to its z...
InWrapX(value: int) -> int — Wrap an x coordinate when x is...
InWrapY(value: int) -> int — Wrap a y coordinate when y is...
InWrapZ(value: int) -> int — Wrap a z coordinate when z is...
__getitem__(key) -> int — Return the integer population count...
__setitem__(key, value: int) — Set the population count at the...

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Full details: API Guide · API Reference

PopGrid

Add(value: int, x: int, y: int=-1, z: int=-1) — Queue a signed population-count...
Update() — Apply queued population-count deltas...
Reset() — Clear population counts and any...
GetPop() -> int — Return the total population summed...
All() -> np.ndarray — Return linear indices of sites whose...

PDEgrid

__len__() -> int — Return the number of lattice sites.
xDim() -> int — Number of lattice sites along x.
yDim() -> int — Number of lattice sites along y;...
zDim() -> int — Number of lattice sites along z;...
nDims() -> int — Number of spatial dimensions.
wrapX() -> bool — Whether x uses periodic boundary...
wrapY() -> bool — Whether y uses periodic boundary...
wrapZ() -> bool — Whether z uses periodic boundary...
Box(x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None, z2: int=None) — Iterate lattice coordinates in the...
Hood(hood, x: int, y: int=None, z: int=None, *, unroll: bool=False) — Apply relative neighborhood offsets...
ToI(x: int, y: int=-1, z: int=-1) -> int — Convert lattice coordinates to a...
ItoX(i: int) -> int — Convert a linear site index to its x...
ItoY(i: int) -> int — Convert a linear site index to its y...
ItoZ(i: int) -> int — Convert a linear site index to its z...
InWrapX(value: int) -> int — Wrap an x coordinate when x is...
InWrapY(value: int) -> int — Wrap a y coordinate when y is...
InWrapZ(value: int) -> int — Wrap a z coordinate when z is...
__getitem__(key) -> float — Return the field value at the...
__setitem__(key, value: float) — Set the field value at the selected...
Add(value: float, x: int, y: int=-1, z: int=-1) — Queue a field-value delta at a site;...
Update() — Apply queued field changes to the...
Reset() — Clear the field and any pending changes.
SetTimeSpaceStep(dt: float, dx: float, dy: float=1.0, dz: float=1.0) — Set the temporal step dt and spatial...
Dx() -> float — Return the configured spatial step...
Dy() -> float — Return the configured spatial step...
Dz() -> float — Return the configured spatial step...
Dt() -> float — Return the configured time step.
Diffusion(rateConstant: float, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None) — Apply scalar-coefficient diffusion...
DiffusionMask(rateConstant: float, mask=None) — Apply diffusion only across...
DiffusionField(rateConstants, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None) — Apply diffusion with spatially...
DiffusionInterfaces(rateConstantsX, rateConstantsY=None, rateConstantsZ=None, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None) — Apply diffusion using rate constants...
DiffusionADI(rateConstant: float, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None) — Apply diffusion using an...
DiffusionRadialCircle(rateConstant: float, outerBC=None) — Apply radial diffusion on a circular...
DiffusionRadialSphere(rateConstant: float, outerBC=None) — Apply radial diffusion on a...
Advection(vx: float, vy: float=0.0, vz: float=0.0, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None) — Apply advection using a constant...
AdvectionField(xVels, yVels=None, zVels=None, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None) — Apply advection using spatially...
AdvectionInterfaces(xVels, yVels=None, zVels=None, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None) — Apply advection using velocity...

PixWindow visualization

IsOpen() -> bool — Return whether the window is...
Update() — Publish the current pixel buffer as...

PixWindow visualization

Save(path: str, block=False) — Save the current pixel buffer to an...
StartGif(path: str, delay=100) — Start recording frames to a GIF at...
AddGifFrame(block=False) — Append the current pixel buffer as a...
StopGif() — Finish GIF recording and write the...
Close() — Close the window and release its...

OpenGLWindow visualization

Update() — Publish the accumulated drawing...
IsOpen() -> bool — Return whether the window is...
StartGif(path, delay=100) — Start recording rendered frames to a...
AddGifFrame(block=False, timeout=30) — Capture the current scene as a GIF...
StopGif(timeout=30) — Finish GIF recording and write the...
Circle(rad, color, x, y, z=None) — Add a circle primitive to the...
Box(xLen, color, x, y, z=None, yLen=None, zLen=None) — Add a box primitive to the current...
BoxSQ(color, x, y, z=None) — Add a unit box centered on the...
Line(width, color, x1, y1, x2, y2, z1=None, z2=None) — Add a line segment to the current frame.
Borders(width, color) — Draw persistent outline borders...
Camera(x, y, z, yaw=None, pitch=None) — Set the 3D camera position and...
Clear() — Clear accumulated drawing primitives...
Background(color) — Set the background color used when...
Save(path: str, block=False) — Save the currently published scene...
Close() — Close the window and release...

Pix drawing buffer

__setitem__(key, color) — Write a packed RGB color to a pixel...
Xdim() — Return the pixel buffer width.
Ydim() — Return the pixel buffer height.
ToI(x, y) — Convert pixel coordinates (x, y) to...
ItoX(i) — Convert a linear pixel index to its...
ItoY(i) — Convert a linear pixel index to its...
__len__() — Return the total number of pixels.

OpenGLDraw primitive collector

Clear() — Discard all primitives accumulated...
Circle(rad, color, x, y, z=0.0) — Add a circle centered at (x, y, z)...
Box(xLen, color, x, y, z=0.0, yLen=-1.0, zLen=-1.0) — Add a box centered at (x, y, z);...
BoxSQ(color, x, y, z=0.0) — Add a unit box centered on lattice...
Line(width, color, x1, y1, x2, y2, z1=0.0, z2=0.0) — Add a line segment between endpoints...